

Problem 1.*Solution:***(a) Interarrival time generation approach:**

- Set $t = 0$, $I = 0$.
- While $t \leq 10$:
 - Generate $U \sim \text{Unif}(0, 1)$.
 - $t = t - \frac{1}{2} \log U$.
 - $I = I + 1$, $A_{(I)} = t$.
- Output $I - 1$ and $A_{(i)}$, $i = 1, 2, \dots, I - 1$, which are the number of arrivals and their arrival times in $[0, 10]$.

Conditional representation approach:

- Generate $N_{(T)} \sim \text{Poi}(\lambda T) = \text{Poi}(20)$.
- Suppose $N_{(T)} = n$. Generate n i.i.d. $\text{Unif}(0, 1)$, say $U_1, \dots, U_n$.
- Sort $U_1, \dots, U_n$ in increasing order to get $U_{(1)} < \dots < U_{(n)}$.
- Output n and $A_i = 10 U_{(i)}$, $i = 1, 2, \dots, n$, which are the number of arrivals and their ordered arrival times in $[0, 10]$.

(b) Interarrival time generation approach for arrivals with a single event-driven logic to track occupancies

- Set $t = 0$.
- Set $\text{busy} = 0$, an empty list (or array) of departure times D (max size = 3).

Step 1: Generate arrivals up to time 10

- While $t \leq 10$:
 - Generate $U \sim \text{Unif}(0, 1)$.
 - $t = t - \frac{1}{2} \log U$.
 - If $t > 10$, exit the loop.
- # Release completed services:
 - * Remove all departure times in D that satisfy departure-time $\leq t$.
 - * For each removed time, decrease busy by 1.
- # Process the new arrival at time t :
 - * If $\text{busy} < 3$:
 - Set $\text{busy} = \text{busy} + 1$.
 - Generate service time $S \sim \text{Exp}(1)$.
 - Set $d = t + S$ and add d to D (this channel's completion time).

* Else (busy = 3): all channels busy $\Rightarrow$ message is lost; do nothing.

Step 2: Determine occupancy at time 10

- After the loop, set $t = 10$.
- Remove from D all departure times ≤ 10 , and for each, decrease busy by 1.
- Now busy = number of channels still busy at time 10.
- Define the indicator

$$I = \begin{cases} 1, & \text{if busy} = 3 \text{ at time } 10, \\ 0, & \text{otherwise.} \end{cases}$$

- Output I .

Replication and estimator

- Run the above independent replication $r = 1, 2, \dots, R$ with $R = 100$.
- Let I_r be the indicator from replication r .
- The estimator of the desired probability is

$$\hat{p} = \frac{1}{R} \sum_{r=1}^R I_r = \frac{1}{100} \sum_{r=1}^{100} I_r.$$

- This $\hat{p}$ is an unbiased Monte Carlo estimate of

$\Pr(\text{all three channels are busy at the 10th hour}).$

Problem 2.

Solution:

(a) Interarrival time generation approach + thinning

Find $\Lambda = 3$ such that $\lambda(t) \leq \Lambda$ for all $t \leq 10$.

- Set $t = 0$, $I = 0$.
- While $t \leq 10$:
 - Generate $U \sim \text{Unif}(0, 1)$.
 - $t = t - \frac{1}{2} \log U$.
 - Generate $V \sim \text{Unif}(0, 1)$.
 - If

$$V \leq \frac{\lambda(t)}{\Lambda} = \frac{2 + \cos(2\pi t)}{3},$$

then accept: $I = I + 1$, $A_{(I)} = t$.

- Output $I - 1$ and $A_{(i)}$, $i = 1, 2, \dots, I - 1$, which are the number of arrivals and their arrival times in $[0, 10]$.

Conditional representation approach and thinning

Find $\Lambda = 3$ such that $\lambda(t) \leq \Lambda$ for all $t \leq 10$.

Initialize $\mathcal{A}$ as an empty set.

- Generate $N_{(T)} \sim \text{Poi}(\Lambda T) = \text{Poi}(30)$.
- Suppose $N_{(T)} = n$. Generate n i.i.d. $\text{Unif}(0, 1)$, say $U_1, \dots, U_n$.
- Sort $U_1, \dots, U_n$ to obtain $U_{(1)} < \dots < U_{(n)}$.
- Set

$$A'_i = T U_{(i)} = 10 U_{(i)}, \quad i = 1, \dots, n.$$

- Generate n i.i.d. $V_i \sim \text{Unif}(0, 1)$.
- For each $i = 1, \dots, n$: If

$$V_i \leq \frac{\lambda(A'_i)}{\Lambda} = \frac{2 + \cos(2\pi A'_i)}{3},$$

then place A'_i into $\mathcal{A}$.

- Sort $\mathcal{A}$ in increasing order, relabel as $A_1 < A_2 < \dots < A_I$.
- Output I and $A_1, \dots, A_I$, which are the number of arrivals and their ordered arrival times in $[0, 10]$.

Modified conditional representation approach

$$\int_0^{10} \lambda(t) dt = \int_0^{10} (2 + \cos(2\pi t)) dt = 20.$$

- Generate

$$N_{(T)} \sim \text{Poi}\left(\int_0^{10} \lambda(t) dt\right) = \text{Poi}(20).$$

- Suppose $N_{(T)} = n$. Generate n i.i.d. variables $V_1, \dots, V_n$ with density

$$f(t) = \frac{\lambda(t)}{\int_0^{10} \lambda(s) ds} = \frac{2 + \cos(2\pi t)}{20}, \quad 0 \leq t \leq 10.$$

ITM: For each i :

- Generate $U_i \sim \text{Unif}(0, 1)$.
- Define the CDF

$$F(t) = \frac{1}{20} \int_0^t (2 + \cos(2\pi x)) dx = \frac{1}{20} \left(2t + \frac{1}{2\pi} \sin(2\pi t) \right).$$

- Numerically solve

$$F(t) = U_i, \quad t \in [0, 10].$$

- Set $V_i = t$ (the numerical solution).

- Sort $V_1, \dots, V_n$ in increasing order to get

$$A_1 < A_2 < \dots < A_n.$$

- Output n and $A_1, A_2, \dots, A_n$, which are the number of arrivals and their ordered arrival times in $[0, 10]$.

- (b) **Performance measure for one replication** In one replication, suppose the simulated arrival times in $[0, 10]$ are

$$0 < A_1 < A_2 < \dots < A_n \leq 10,$$

and no arrivals after time 10 are included.

For a counting process that jumps by +1 at each A_i , one can show

$$\int_0^{10} N(t) dt = \sum_{i=1}^n (10 - A_i).$$

So for the replication, define

$$Y = \frac{1}{10} \sum_{i=1}^n (10 - A_i).$$

Replication and estimator (for each approach)

For replication $r = 1, 2, \dots, 100$:

- Use one of the three methods in part (a) to simulate a path on $[0, 10]$, obtaining arrival times

$$A_1^{(r)} < A_2^{(r)} < \dots < A_{n_r}^{(r)}.$$

- Compute

$$Y_r = \frac{1}{10} \sum_{i=1}^{n_r} (10 - A_i^{(r)}).$$

After 100 replications, the estimator is

$$\hat{\theta} = \frac{1}{100} \sum_{r=1}^{100} Y_r.$$

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1  Problem 2(b) estimates:
2
3  Method 1 (Interarrival + thinning):  9.78569217471732
4  Method 2 (Conditional + thinning):  9.67194256037633
5  Method 3 (Modified conditional):  10.111998213374257

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Problem 3.

Solution: **Interarrival times** Interarrival times X_n are i.i.d. $\text{Exp}(1/2)$, rate $= \frac{1}{2}$, so

$$X = -2 \log U, \quad U \sim \text{Unif}(0, 1).$$

Service times Service times S_n are i.i.d. $\text{Gamma}(3, 2)$, and can be written as

$$S_n = Y_1 + Y_2 + Y_3,$$

where $Y_k \sim \text{Exp}(2)$, and

$$Y = -\frac{1}{2} \log U.$$

Lindley recursion Waiting times satisfy Lindley's recursion:

$$W_{n+1} = \max\{0, W_n + S_n - X_{n+1}\}, \quad W_0 = 0, S_0 = 0.$$

Pseudo-code

- Total number of customers simulated: $N = 1000$. Burn-in: $B = 100$.
- Set $W_0 = 0, S_0 = 0, \text{sum_wait} = 0, \text{count} = 0$.
- For $n = 0, 1, 2, \dots, N - 1$:
 - Generate interarrival time $X_{n+1} \sim \text{Exp}(1/2)$: Generate $U \sim \text{Unif}(0, 1)$ and set $X_{n+1} = -2 \log U$.
 - Compute the waiting time of customer $n + 1$:

$$W_{n+1} = \max\{0, W_n + S_n - X_{n+1}\}.$$

- Generate service time $S_{n+1} \sim \text{Gamma}(3, 2)$ as sum of 3 exponentials:

$$\text{For } k = 1, 2, 3: \quad U_k \sim \text{Unif}(0, 1), \quad Y_k = -\frac{1}{2} \log U_k.$$

$$S_{n+1} = Y_1 + Y_2 + Y_3.$$

- If $n + 1 > B$, then:
 - * `sum_wait` $\leftarrow$ `sum_wait` + W_{n+1}
 - * `count` $\leftarrow$ `count` + 1

Estimator The estimator of the long-run average waiting time is

$$\hat{w} = \frac{\text{sum_wait}}{\text{count}}.$$

Report $\hat{w}$ as the simulated long-run average waiting time of customers in the $G/G/1$ queue.

1 long-run average waiting time: 3.2167959966456006

Problem 4.

Solution:

(a) We want to estimate

$$p = P(X > 10), \quad X \sim \text{Gamma}(2, 1).$$

The original density is

$$f(x) = xe^{-x}, \quad x > 0.$$

We choose an importance sampling proposal distribution

$$Y \sim \text{Gamma}(\alpha, 1),$$

and tune α so that the mean of Y equals 10. Since a $\text{Gamma}(\alpha, 1)$ distribution has mean

$$E[Y] = \frac{\alpha}{1} = \alpha,$$

we must set

$$\alpha = 10.$$

Thus the proposal is

$$Y \sim \text{Gamma}(10, 1),$$

with density

$$g(x) = \frac{1}{\Gamma(10)} x^9 e^{-x}.$$

The likelihood ratio (Radon–Nikodym derivative) is

$$L(x) = \frac{f(x)}{g(x)} = \frac{xe^{-x}}{\frac{1}{\Gamma(10)}x^9e^{-x}} = \Gamma(10)x^{-8} = 9!x^{-8}.$$

We are estimating

$$p = P(X > 10) = E[\mathbf{1}_{\{X > 10\}}],$$

so under importance sampling,

$$\hat{p} = \mathbf{1}_{\{Y > 10\}} L(Y) = \mathbf{1}_{\{Y > 10\}} (9!) Y^{-8}.$$

Thus, one replication produces:

$$Y \sim \text{Gamma}(10, 1)$$

$$\hat{p} = \mathbf{1}_{\{Y > 10\}} (9!) Y^{-8}.$$

(b) **Exponential tilting** The original density is

$$f(x) = xe^{-x}, \quad x > 0.$$

By exponential tilting with parameter θ , the tilted density is

$$f_\theta(x) = \frac{e^{\theta x} f(x)}{\mathbb{E}[e^{\theta X}]} = x(1 - \theta)^2 e^{-(1-\theta)x}.$$

Thus under tilting,

$$X_\theta \sim \text{Gamma}(2, 1 - \theta).$$

Since

$$\mathbb{E}[X_\theta] = \frac{2}{1 - \theta},$$

we set this equal to 10 to match the mean:

$$\frac{2}{1 - \theta} = 10 \quad \Rightarrow \quad 1 - \theta = 0.2 \quad \Rightarrow \quad \theta = 0.8.$$

Therefore the tilted proposal distribution is

$$X_\theta \sim \text{Gamma}(2, 0.2).$$

The likelihood ratio is

$$L(x) = \frac{f(x)}{f_\theta(x)} = \frac{xe^{-x}}{x(0.2)^2 e^{-0.2x}} = (0.2)^{-2} e^{-0.8x}.$$

We want

$$p = P(X > 10) = \mathbb{E}[\mathbf{1}_{\{X > 10\}}].$$

Under importance sampling,

$$\hat{p} = \mathbf{1}_{\{X_\theta > 10\}} L(X_\theta) = \mathbf{1}_{\{X_\theta > 10\}} (0.2)^{-2} e^{-0.8X_\theta}.$$

One replication produces:

- Sample $X_\theta \sim \text{Gamma}(2, 0.2)$,
- Output

$$\hat{p} = \mathbf{1}_{\{X_\theta > 10\}} (0.2)^{-2} e^{-0.8X_\theta}.$$

Legitimacy of a shifted gamma proposal Consider a shifted gamma proposal of the form

$$Y = X + 10, \quad X \sim \text{Gamma}(2, 1).$$

This proposal is *not* legitimate for importance sampling.

The reason is that the shifted distribution has support

$$\text{supp}(Y) = (10, \infty),$$

while the original gamma distribution has support

$$\text{supp}(f) = (0, \infty).$$

Thus for all $x < 10$,

$$f(x) > 0 \quad \text{but} \quad g(x) = 0,$$

which means the likelihood ratio

$$\frac{f(x)}{g(x)}$$

is not defined.

Therefore a shifted gamma cannot be used as an importance sampling proposal.

Problem 5.

Solution:

- (a) We consider an $M/M/1$ queue with arrival rate 2 per minute and exponential service times with mean 1 minute. Let T_i denote the total time that customer i spends in the system. We wish to estimate

$$\mu = \mathbb{E}[T_1 + T_2 + \cdots + T_{10}] = \mathbb{E}[Y], \quad Y = \sum_{i=1}^{10} T_i.$$

Waiting times satisfy Lindley's recursion

$$W_{n+1} = \max\{0, W_n + S_n - X_{n+1}\}, \quad T_{n+1} = W_{n+1} + S_{n+1},$$

with $W_0 = 0$, $S_0 = 0$, and where X_{n+1} is the interarrival time between customers n and $n + 1$.

Arrivals follow a Poisson process with rate 2 per minute, so interarrival times are $X_i \sim \text{Exp}(2)$. Service times are exponential with mean 1, so $S_i \sim \text{Exp}(1)$.

For one replication:

- Set $W_0 = 0$, $S_0 = 0$.
- For $i = 1, 2, \dots, 11$: generate $X_i \sim \text{Exp}(2)$ via $X_i = -\frac{1}{2} \log U$, $U \sim \text{Unif}(0, 1)$.
- For $i = 1, 2, \dots, 10$: generate $S_i \sim \text{Exp}(1)$ via $S_i = -\log U'$, $U' \sim \text{Unif}(0, 1)$.
- For $n = 0, 1, \dots, 9$:
 - Compute

$$W_{n+1} = \max\{0, W_n + S_n - X_{n+1}\}.$$

– Compute

$$T_{n+1} = W_{n+1} + S_{n+1}.$$

• Compute the total

$$Y = \sum_{i=1}^{10} T_i.$$

Over $R = 100$ replications:

- For $r = 1, 2, \dots, R$, run the above procedure and obtain $Y^{(r)}$.
- The naive Monte Carlo estimator is

$$\hat{\mu}_{\text{naive}} = \frac{1}{R} \sum_{r=1}^R Y^{(r)} = \frac{1}{100} \sum_{r=1}^{100} Y^{(r)}.$$

- The estimated variance per replication is

$$\hat{\sigma}_{\text{naive}}^2 = \frac{1}{R-1} \sum_{r=1}^R (Y^{(r)} - \hat{\mu}_{\text{naive}})^2 = \frac{1}{99} \sum_{r=1}^{100} (Y^{(r)} - \hat{\mu}_{\text{naive}})^2.$$

(b) Control variate: Define the control variate

$$C_1^{(r)} = \sum_{i=1}^{10} S_i^{(r)},$$

since each $S_i \sim \text{Exp}(1)$ with mean 1, and therefore

$$\mathbb{E}[C_1] = 10.$$

For one replication:

- Set $W_0 = 0$, $S_0 = 0$.
- For $i = 1, 2, \dots, 11$: generate

$$X_i \sim \text{Exp}(2), \quad X_i = -\frac{1}{2} \log U, \quad U \sim \text{Unif}(0, 1).$$

- For $i = 1, 2, \dots, 10$: generate

$$S_i \sim \text{Exp}(1), \quad S_i = -\log U', \quad U' \sim \text{Unif}(0, 1).$$

- For $n = 0, 1, \dots, 9$:

– Compute

$$W_{n+1} = \max\{0, W_n + S_n - X_{n+1}\}.$$

– Compute

$$T_{n+1} = W_{n+1} + S_{n+1}.$$

- Compute the target quantity

$$Y = \sum_{i=1}^{10} T_i.$$

- Compute the control variate

$$C_1 = \sum_{i=1}^{10} S_i.$$

Over $R = 100$ replications:

- For $r = 1, 2, \dots, R$, collect the pair

$$(Y^{(r)}, C_1^{(r)}).$$

- Compute sample means:

$$\bar{Y} = \frac{1}{R} \sum_{r=1}^R Y^{(r)}, \quad \bar{C}_1 = \frac{1}{R} \sum_{r=1}^R C_1^{(r)}.$$

- Compute sample covariance and variance:

$$S_{Y,C_1} = \frac{1}{R-1} \sum_{r=1}^R (Y^{(r)} - \bar{Y})(C_1^{(r)} - \bar{C}_1),$$

$$S_{C_1}^2 = \frac{1}{R-1} \sum_{r=1}^R (C_1^{(r)} - \bar{C}_1)^2.$$

- Set the optimal control coefficient:

$$b_1 = \frac{S_{Y,C_1}}{S_{C_1}^2}.$$

- For each replication, define the controlled value:

$$Z^{(r)} = Y^{(r)} - b_1(C_1^{(r)} - 10).$$

- Point estimate:

$$\hat{\mu}_{CV1} = \frac{1}{R} \sum_{r=1}^R Z^{(r)}.$$

- Estimated variance per replication:

$$\hat{\sigma}_{CV1}^2 = \frac{1}{R-1} \sum_{r=1}^R (Z^{(r)} - \hat{\mu}_{CV1})^2.$$

Compare $\hat{\sigma}_{\text{naive}}^2$ and $\hat{\sigma}_{\text{CV1}}^2$. The variance reduction factor is

$$\frac{\hat{\sigma}_{\text{naive}}^2}{\hat{\sigma}_{\text{CV1}}^2}.$$

Since the total time in system Y is positively correlated with the total service time C_1 , the control variate should noticeably reduce variance.

(c) Control variate: Define the control variate

$$C_2^{(r)} = \sum_{i=1}^{10} S_i^{(r)} - \sum_{i=1}^{10} X_{i+1}^{(r)}.$$

Since $\mathbb{E}[S_i] = 1$ and $\mathbb{E}[X_{i+1}] = \frac{1}{2}$, we have

$$\mathbb{E}[C_2] = 10 - 5 = 5.$$

For one replication:

- Set $W_0 = 0, S_0 = 0$.
- For $i = 1, 2, \dots, 11$: generate

$$X_i \sim \text{Exp}(2), \quad X_i = -\frac{1}{2} \log U, \quad U \sim \text{Unif}(0, 1).$$

- For $i = 1, 2, \dots, 10$: generate

$$S_i \sim \text{Exp}(1), \quad S_i = -\log U', \quad U' \sim \text{Unif}(0, 1).$$

- For $n = 0, 1, \dots, 9$:

– Compute

$$W_{n+1} = \max\{0, W_n + S_n - X_{n+1}\}.$$

– Compute

$$T_{n+1} = W_{n+1} + S_{n+1}.$$

- Compute

$$Y = \sum_{i=1}^{10} T_i, \quad C_2 = \sum_{i=1}^{10} S_i - \sum_{i=1}^{10} X_{i+1}.$$

Over $R = 100$ replications:

- For $r = 1, 2, \dots, R$, collect the pair

$$(Y^{(r)}, C_2^{(r)}).$$

- Compute sample means:

$$\bar{Y} = \frac{1}{R} \sum_{r=1}^R Y^{(r)}, \quad \bar{C}_2 = \frac{1}{R} \sum_{r=1}^R C_2^{(r)}.$$

- Compute sample covariance and variance:

$$S_{Y,C_2} = \frac{1}{R-1} \sum_{r=1}^R (Y^{(r)} - \bar{Y})(C_2^{(r)} - \bar{C}_2),$$

$$S_{C_2}^2 = \frac{1}{R-1} \sum_{r=1}^R (C_2^{(r)} - \bar{C}_2)^2.$$

- Set the optimal control coefficient:

$$b_2 = \frac{S_{Y,C_2}}{S_{C_2}^2}.$$

- For each replication, define the controlled value:

$$Z^{(r)} = Y^{(r)} - b_2 (C_2^{(r)} - 5).$$

- Point estimate:

$$\hat{\mu}_{CV2} = \frac{1}{R} \sum_{r=1}^R Z^{(r)}.$$

- Estimated variance per replication:

$$\hat{\sigma}_{CV2}^2 = \frac{1}{R-1} \sum_{r=1}^R (Z^{(r)} - \hat{\mu}_{CV2})^2.$$

Quality comparison. Compare the three estimated variances:

$$\hat{\sigma}_{\text{naive}}^2, \quad \hat{\sigma}_{CV1}^2, \quad \hat{\sigma}_{CV2}^2.$$

Because the recursion involves the difference $S_n - X_{n+1}$, the quantity C_2 is expected to be more strongly correlated with Y than C_1 .

Therefore, in practice we expect

$$\hat{\sigma}_{CV2}^2 \leq \hat{\sigma}_{CV1}^2 \ll \hat{\sigma}_{\text{naive}}^2.$$

```

1  (a) Naive estimator:
2      Mean estimate = 34.05723441214564
3      Variance per replication = 314.0277865209554
4
5  (b) Control variate C1 = sum S_i:
6      Mean estimate = 35.5829113419057
7      Variance per replication = 114.45185154972575
8      b1 = 5.086470204033914
9      Variance reduction = 2.7437545331848137
10
11 (c) Control variate C2 = sum S_i - sum X_{i+1}:
12     Mean estimate = 34.8143517343161
13     Variance per replication = 71.08758409199845
14     b2 = 4.576373981630326
15     Variance reduction = 4.417477264588909

```

Problem 6.*Solution:*(a) Naive Monte Carlo: **One replication:**

- Sample $P \sim \text{Beta}(1, 19)$: generate $U \sim \text{Unif}(0, 1)$, set

$$P = (1 - U)^{1/19}. \quad (\text{inverse transform})$$

- For $n = 1, 2, \dots, 100$:

- Sample $D_n \sim \text{Bernoulli}(P)$: generate $U' \sim \text{Unif}(0, 1)$, set

$$D_n = \begin{cases} 1, & U' < P, \\ 0, & \text{otherwise.} \end{cases}$$

- Sample $X_n \sim N(3, 1)$.

- If a normal RNG is not available: generate 12 independent uniforms $U''_1, \dots, U''_{12} \sim \text{Unif}(0, 1)$, form

$$Z = \sum_{i=1}^{12} U''_i - 6,$$

which has mean 0, variance 1 (Irwin–Hall approximation), then set

$$X_n = 3 + Z.$$

- Compute the total loss

$$L = \sum_{n=1}^{100} D_n X_n.$$

- Record indicator

$$Y = \mathbf{1}_{\{L > 45\}}.$$

Estimator over R replications:

- For $r = 1, 2, \dots, R$, run the above and obtain $Y^{(r)}$.
- The naive MC estimator is

$$\hat{\alpha}_{\text{naive}} = \frac{1}{R} \sum_{r=1}^R Y^{(r)}.$$

- Estimated variance per replication:

$$\hat{\sigma}_{\text{naive}}^2 = \frac{1}{R-1} \sum_{r=1}^R (Y^{(r)} - \hat{\alpha}_{\text{naive}})^2.$$

(b) Conditional Monte Carlo: The conditional Monte Carlo estimator is

$$\alpha = \mathbb{E}[P(L > 45)] = \mathbb{E}[\phi(P)],$$

where

$$\phi(p) = P(L > 45 \mid P = p).$$

Given $P = p$, the number of defaults

$$K \sim \text{Bin}(100, p).$$

Given $K = k$, the loss is

$$L = \sum_{i=1}^k X_i, \quad X_i \stackrel{\text{i.i.d.}}{\sim} N(3, 1).$$

Thus

$$L \mid K = k \sim N(3k, k).$$

Therefore

$$P(L > 45 \mid K = k) = 1 - \Phi\left(\frac{45 - 3k}{\sqrt{k}}\right),$$

with the convention that if $k = 0$, the probability is 0.

One replication:

- Sample $P \sim \text{Beta}(1, 19)$ using inverse transform: generate $U \sim \text{Unif}(0, 1)$ and set

$$P = (1 - U)^{1/19}.$$

- Compute

$$p^{(r)} = \sum_{k=0}^{100} \binom{100}{k} P^k (1 - P)^{100-k} \left[1 - \Phi\left(\frac{45 - 3k}{\sqrt{k}}\right) \right],$$

with the term defined as 0 for $k = 0$.

Estimator over R replications:

- For $r = 1, 2, \dots, R$, run the above and obtain $p^{(r)}$.
- The conditional MC estimator is

$$\hat{\alpha}_{\text{cml}} = \frac{1}{R} \sum_{r=1}^R p^{(r)}.$$

- Estimated variance per replication:

$$\hat{\sigma}_{\text{cml}}^2 = \frac{1}{R-1} \sum_{r=1}^R (p^{(r)} - \hat{\alpha}_{\text{cml}})^2.$$

- (c) Comments: Naive Monte Carlo suffers from large variance because the event $L > 45$ is relatively rare. Conditional Monte Carlo dramatically reduces variance because it removes the Bernoulli randomness from the indicators D_n and converts the inner simulation into a closed-form Gaussian tail probability conditioned on K . Thus conditional Monte Carlo produces far lower variance and much more efficient estimates.

```

1  === Problem 6 ===
2  (1) Naive Monte Carlo
3      Estimate = 0.04
4      Variance per replication = 0.03878787878787879
5
6  (2) Conditional Monte Carlo
7      Estimate = 0.043022483599930474
8      Variance per replication = 0.017063530783707277
9
10 Variance reduction factor = 2.2731449475224967

```